

Mahmoud Ayman

Backend Developer — Software Engineer

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SUMMARY

Computer Science graduate with a strong foundation in Operating Systems, Databases, Networking, Data Structures & Algorithms, and Software Engineering. Solved 300+ algorithmic problems across LeetCode, NeetCode, and Codeforces before specializing in backend development using ASP.NET Core. I enjoy becoming a T-shaped engineer by combining deep backend knowledge with frontend experience using React. Currently preparing for the AWS Certified Solutions Architect – Associate (SAA-C03) certification and actively contributing to open-source projects including Checkstyle and pgwatch.

EDUCATION

Beni Suef University

B.Sc. in Computer Science, Faculty of Computers and Artificial Intelligence

Beni Suef, Egypt

Sep. 2022 – Aug. 2026

- **GPA:** 3.25 / 4.00
- **Relevant Coursework:** Data Structures, Algorithm Design, Software Engineering, Operating Systems, Database Internals, Networking, Artificial Intelligence, Machine Learning, Compiler Construction

PROJECTS

Quiz Monitor | [GitHub](#) | [Graduation Project](#)

ASP.NET Core, PostgreSQL, EF Core, Docker

Built the backend for an AI-driven exam proctoring platform. Designed an N-Tier architecture to support both student web interfaces and instructor mobile applications.

- Developed a RESTful API using ASP.NET Core 9 and Entity Framework Core to manage the full exam lifecycle, from instructor creation to secure student submission.
- Designed the PostgreSQL database schema on Supabase. Used Unit of Work and Repository patterns to keep data consistent during complex operations.
- Built a logging system to record AI-detected cheating events in real-time, such as eye-tracking anomalies and unauthorized tab switching.
- Containerized the application with a multi-stage Dockerfile to reduce image size, and deployed it to Railway.

MockAPIs | [GitHub](#) | [Live Demo](#)

ASP.NET Core, React, PostgreSQL, Docker, AWS, Railway, Vercel

Built a full-stack platform that lets frontend teams design custom schemas and generate mock REST APIs. Created a 3-layer .NET backend and a React frontend, and designed an AWS architecture based on the AWS Well-Architected Framework.

- Built management APIs and a React interface to configure resources, custom fields, dynamic endpoints, and automated fake data generation.
- Designed an AWS architecture with EC2 Auto Scaling, Amazon RDS PostgreSQL, CloudFront/S3 for hosting, and AWS WAF for security.
- Fixed an in-memory pagination bottleneck by moving the logic to PostgreSQL. Tested with k6 (50 concurrent users, 10K records), which reduced latency by 98.4% (859ms to 13.5ms) and increased throughput by ~80%.
- Containerized the app with Docker and deployed the live prototype to Railway (API), Supabase (Postgres), and Vercel (React).

AskFm Full-Stack Platform | [GitHub](#) | [Demo Video](#) *ASP.NET Core, EF Core, SignalR, SQL Server, React, Vite, xUnit*

Built a full-stack social Q&A platform with a .NET backend and a React frontend featuring real-time notifications via SignalR. Used N-Tier architecture with Repository and Unit of Work patterns for maintainability.

- Developed a responsive React frontend with context-based state management, JWT authentication flow, and live SignalR event listeners for instant UI updates.
- Built a real-time notification system using SignalR to push user interaction events directly to active web clients without page refresh.
- Fixed N+1 query problems by refactoring the data access layer to use compiled IQueryable projections and batched lookups.
- Reduced API execution time by over 99% (from > 30s to < 15ms) during stress tests with 15,000+ records by avoiding heavy object creation.
- Optimized bulk database updates using EF Core `ExecuteUpdateAsync`, which saved memory and reduced database round-trips.

pgwatch Copilot Demo | [GitHub](#) | [Demo Video](#)

Golang, PostgreSQL, CLI

pgwatch is an open-source PostgreSQL monitoring tool. I built a CLI on top of it that turns plain-English questions into safe PostgreSQL queries, returning results as table/JSON/CSV.

- Built a query pipeline in Go (question → schema inspect → SQL generation → validation → execution) so operators can get metrics without writing SQL.
- Added a safety layer that parses the generated SQL, blocks risky functions, and only allows read-only queries with timeouts to prevent destructive actions.
- Added live schema introspection to discover tables and JSONB keys, which improves the LLM's ability to write accurate queries.
- Configured the LLM (Gemini) for deterministic behavior, added a dry-run mode, and wrote unit tests for SQL validation. Created demo data to make testing reproducible.

MechanicShop API | [GitHub](#)

C#, ASP.NET Core 9, EF Core, PostgreSQL, k6

A RESTful API for an auto repair shop to handle repair tasks, inventory management, and automated invoicing.

- Built the backend using ASP.NET Core 9 and Clean Architecture to keep the code organized, maintainable, and easy to test.
- Optimized database queries by moving pagination, filtering, and joins to PostgreSQL. This reduced average API response time by 98.9% (from 2.72s to 28ms) and increased throughput by 365% in k6 load tests.
- Designed the database model to keep historical pricing data and used the Unit of Work pattern to ensure data consistency during invoice generation.
- Secured the API with JWT authentication and role-based access control (RBAC).

TECHNICAL SKILLS

Languages: C#, Golang, SQL, C++, Java, Python, JavaScript

Frameworks: ASP.NET Core, Entity Framework Core, SignalR, React, LINQ

Databases: PostgreSQL, SQL Server

Tools: Docker, Git, GitHub, Linux, AWS, VS Code, Visual Studio

Concepts: REST APIs, Clean Architecture, SOLID Principles, Design Patterns, Data Structures, Algorithms

COURSES & TRAINING

Mastering ASP.NET APIs | [Link](#)

Online

Completed coursework and quizzes focusing on advanced API development, routing, and middleware in ASP.NET Core.

AWS Solutions Architect Associate (SAA) Preparation - Manara | [Link](#)

Online

Completed training and quizzes covering core AWS services, architecture design, and the Well-Architected Framework.